Exponential online mistakes after an arbitrarily small jump is added to tanh

ChatGPT-6 Astra Kaarel Hänni Dmitry Vaintrob*

6 November 2026

Abstract

For every fixed $\epsilon > 0$, consider the bounded activation $\phi_\epsilon(t) = \tanh(t) + \epsilon \operatorname{sign}(t)$. Its uniform distance from tanh is ϵ, but it has a jump at zero. We construct a shallow Bayesian neural-network learner with the standard coordinate-isotropic Gaussian prior that makes exponentially many online mistakes, at every hidden width in its strong-prior limit. The target has a Boolean circuit of size $\Theta(k)$ and a teacher of width $\lceil 1/\epsilon \rceil$ whose weights are bounded by one. A fixed input permutation begins with $O(k^2)$ calibration examples and has $2^k - k - 1$ designated errors, with $O(k)$ intervening examples per error. The prior is conditioned on absolute output margin greater than $1/2$ on the entire Boolean cube, and then on exact training labels. The proof gives an exact Gaussian energy minimum from an affine margin certificate, controls every nonlinear competitor, and obtains a width-uniform energy gap using a compact relaxation over neuron distributions. Direct width-dependent prior limits are also covered. For the same learner, a separate energy compression argument gives IID sample complexity $O_\epsilon((d^3 + \log(1/\beta))/\eta)$ for test confidence greater than $2/3$ outside input mass η. The result uses a fixed nonzero jump; it does not settle the question for ordinary tanh or any continuous approximation to this activation.

1 The model and main result

We study an activation arbitrarily close to tanh in the uniform norm and prove an all-width online lower bound for it. The activation is

$$\phi_\epsilon(t) = \tanh(t) + \epsilon \operatorname{sign}(t), \qquad \epsilon > 0, \qquad \operatorname{sign}(0) = 0.$$

The coefficient ϵ is fixed across input dimensions and learner widths. The activation is odd, increasing, and bounded by $M = 1 + \epsilon$ in absolute value. Its distance from ordinary tanh in the uniform norm is exactly ϵ.

For $x \in \{-1, 1\}^d$, put $\bar{x} = (1, x)$. A width-N learner has one hidden layer and output

$$F(x) = \sum_{j=1}^{N} v_j \phi_\epsilon(u_j \cdot \bar{x}), \qquad u_{ji} \sim \mathcal{N}(0, N\sigma^2/d), \qquad v_j \sim \mathcal{N}(0, \sigma^2/N), \tag{1}$$

with all Gaussian coordinates independent. The coefficient of the constant input has the same variance as the other incoming weights. There is no anisotropy or additional readout penalty.

The initial prior is conditioned on $|F(x)| > 1/2$ at every cube input. After labels are revealed, it is further conditioned on $f(x)F(x) > 1/2$ at those inputs. A fresh posterior sample makes an

*Authors are listed in alphabetical order.

error when it assigns the wrong sign. We also consider posterior-label majority and the convention requiring correct-label posterior probability greater than 2/3.

At fixed N, the strong-prior limit is $\sigma \downarrow 0$. For direct width limits, set

$$\sigma_N^2 = \tau^2 N^{-\gamma}, \qquad \mathrm{Var}(u_{ji}) = \tau^2 N^{1-\gamma}/d, \qquad \mathrm{Var}(v_j) = \tau^2 N^{-\gamma-1}, \tag{2}$$

where $\tau > 0$ and $\gamma \geq 1$ are fixed.

Theorem 1. *Fix $\epsilon > 0$. Let $k \geq 8$, $d = 2k + 2$, and assume $d \geq 4\max\{1, \epsilon^{-6}\}$. Write the inputs as (x, y, z, s) with $x, y \in \{-1, 1\}^k$ and $z, s \in \{-1, 1\}$. Put $A = 2^k$ and define*

$$f_k(x, y, z, s) = \mathrm{sign}\left(-(A-1)s - Az + \sum_{i=0}^{k-1} 2^i (x_i - y_i)\right). \tag{3}$$

There is a fixed permutation of the full input cube, depending only on k, with these properties.

1. *The target has Boolean circuit size $\Theta(k)$ and a teacher of hidden width $\lceil 1/\epsilon \rceil$, with all weights bounded by one and signed output greater than one everywhere.*

2. *A calibration prefix of at most $2k^2 + 6k + 2$ examples is followed by $T_k = 2^k - k - 1 \geq 2^{k-1}$ designated queries. There are at most $4k$ examples in each query block, including its designated first query. At every fixed width $N \geq 1$, the posterior wrong-label probability at each designated query tends to one as $\sigma \downarrow 0$.*

3. *For every fixed $\gamma > 1$ and $\tau > 0$ in (2), those same wrong-label probabilities tend to one as $N \to \infty$.*

4. *For every $\delta \in (0, 1)$ there exists $\tau_0(k, \epsilon, \delta) > 0$ such that, for all $0 < \tau \leq \tau_0$, all $\gamma \geq 1$, and all $N \geq 1$, posterior sampling makes every designated mistake with probability at least $1 - \delta$. Each designated posterior majority is also wrong when δ is sufficiently small, and the 2/3 correctness criterion fails there.*

5. *In the same cold and direct-width regimes, the learner has IID sample complexity $O_\epsilon((d^3 + \log(1/\beta))/\eta)$ for correct-label posterior confidence greater than 2/3 outside a set of input mass at most η, with sample confidence at least $1 - \beta$. The precise variance and width quantifiers are given in Theorem 7.*

The comparison includes every nonlinear network at every width; no choice among tied minimum-energy predictors is made.

The teacher-width bound is independent of k at fixed activation. The dimension restriction grows as ϵ decreases. Neither this restriction nor the prior limits can be interchanged with $\epsilon \downarrow 0$ to obtain a theorem for ordinary tanh.

2 An exact energy reduction under a covariance certificate

Consider a nonempty linearly separable labeled set, and put $z_i = y_i \bar{x}_i$. Let a be its minimum-norm affine unit-margin separator and $Q = \|a\|^2$. Suppose there are probabilities ρ_i on active constraints such that

$$z_i \cdot a = 1 \quad (\rho_i > 0), \qquad \sum_i \rho_i z_i = a/Q, \qquad C := \sum_i \rho_i z_i z_i^T \preceq 2I. \tag{4}$$

The first two conditions are the usual normalized optimality certificate. The covariance condition is an additional assumption; it will be verified explicitly for the histories used below. It also implies $Ca = a/Q$.

Write $b_j = Nv_j$. The energy and output are

$$E_N = \frac{1}{N}\sum_j (d\|u_j\|^2 + b_j^2), \qquad F(x) = \frac{1}{N}\sum_j b_j \phi_\epsilon(u_j \cdot \bar{x}). \tag{5}$$

Theorem 2 (Exact energy and equality)**.** *Assume (4) and $d \geq 4\max\{1, \epsilon^{-6}\}$. At every hidden width $N \geq 1$, the minimum energy over networks satisfying the training margins $y_i F(x_i) \geq 1/2$ is*

$$J_d(Q) = \min_{r \geq 0} \left\{ dQr^2 + \frac{1}{4(\epsilon + \tanh r)^2} \right\}. \tag{6}$$

The unique minimizing radius is positive and satisfies

$$4dQr(\epsilon + \tanh r)^3 = \operatorname{sech}^2 r. \tag{7}$$

The profile is strictly increasing in Q and is below $1/(4\epsilon^2)$. It is attained by identical neurons

$$u_j = ra, \qquad b_j = \frac{1}{2(\epsilon + \tanh r)}.$$

At equality every neuron has this form or has both signs reversed. Consequently the entire output function is unique. The same assertions hold for arbitrary probability distributions over neurons in place of the uniform distribution on N neurons.

Proof. Put $h = \epsilon + \tanh r$. The scalar objective has derivative $2dQr - \operatorname{sech}^2 r/(2h^3)$, which is strictly increasing from a negative value to a positive value. This proves the existence and uniqueness of the positive optimum and (7). Define

$$\beta = \operatorname{sech}^2 r, \qquad \alpha = \epsilon + \tanh r - r\beta, \qquad \lambda = h^{-2}, \qquad K = \lambda^2/4, \qquad M = 1 + \epsilon.$$

Then $0 < \alpha \leq M$, $0 < \beta \leq 1$, $h = \alpha + \beta r$, and $K \leq 1/(4\epsilon^4)$. The inequality $\alpha \geq \epsilon$ follows by differentiating $\tanh r - r\operatorname{sech}^2 r$, whose derivative is nonnegative. Stationarity gives $dQr = K\beta h$, so

$$dQt^2 - K(\alpha + \beta t)^2 \geq dQr^2 - Kh^2 =: \zeta = -\frac{\alpha}{4h^3} < 0 \quad (t \geq 0). \tag{8}$$

Indeed its leading coefficient is $dQ - K\beta^2 = K\alpha\beta/r > 0$ and its minimum is at $t = r$.

For a neuron u, decompose $u = ta + w$ with $w \perp a$, and regard $s = z_i \cdot u$ as a random variable under ρ. The active identities imply

$$\mathbb{E}_\rho s = t, \qquad \operatorname{Var}_\rho(s) = w^T C w \leq 2\|w\|^2.$$

Since $\sup_s |\tanh'' s| < 1$, Taylor's theorem gives

$$|\mathbb{E}_\rho \tanh s - \tanh t| \leq \tfrac{1}{2}\operatorname{Var}_\rho(s) \leq \|w\|^2.$$

For $S(u) = \mathbb{E}_\rho \phi_\epsilon(s)$, the threshold part has absolute average at most ϵ. Concavity of tanh on the positive half-line therefore yields

$$|S(u)| \leq \epsilon + |\tanh t| + \|w\|^2 \leq \alpha + \beta|t| + \|w\|^2. \tag{9}$$

Also $|S(u)| \le M$.

For any readout b, complete the square:

$$d\|u\|^2 + b^2 - \lambda b S(u) \ge d\|u\|^2 - KS(u)^2. \tag{10}$$

If $\|u\|^2 \ge KM^2/d$, the right side is nonnegative and hence strictly greater than ζ. Otherwise set $q = \|w\|^2$ and $R_0 = \sqrt{K}M/\sqrt{d}$. Then $q \le R_0^2$ and $|t| \le R_0/\sqrt{Q}$. By (9), the right side of (10) is at least

$$[dQt^2 - K(\alpha + \beta|t|)^2] + q[d - 2K(\alpha + \beta|t|) - Kq]. \tag{11}$$

Any unit-margin constraint gives $Q \ge 1/(d+1)$, so $dQ \ge 1/2$. Consequently

$$2K(\alpha + \beta|t|) \le 2KM(1 + \sqrt{2K}).$$

If $0 < \epsilon \le 1$, this is at most $\epsilon^{-4}(1 + 1/(\sqrt{2}\epsilon^2)) \le 2\epsilon^{-6} \le d/2$. If $\epsilon \ge 1$, use $M \le 2\epsilon$ to bound it by $\epsilon^{-3}(1 + 1/(\sqrt{2}\epsilon^2)) \le 2 \le d/2$. Similarly,

$$Kq \le K^2M^2/d \le \begin{cases} 1/(4d\epsilon^8) \le d/64, & 0 < \epsilon \le 1, \\ 1/(4d\epsilon^6) \le d/64, & \epsilon \ge 1. \end{cases}$$

Thus the second bracket in (11) is at least $31d/64 > 0$. Equations (8)–(11) prove the atomwise floor ζ for every u, b.

Averaging over neurons and using the signed training margins gives

$$E_N \ge \lambda \mathbb{E}_j[b_j S(u_j)] + \zeta \ge \lambda/2 + \zeta = dQr^2 + \frac{1}{4h^2}.$$

The cloned construction attains equality: every signed training preactivation is at least r, and active ones equal r. The same proof integrates over any neuron probability law with finite energy.

Equality forces $w = 0$, $|t| = r$, and equality in square completion. Thus $u = \pm ra$ and $b = \pm 1/(2h)$ with matching signs. Oddness of the activation makes all such atoms have the same output. Finally, the negative derivative at zero puts the optimum below $1/(4\epsilon^2)$. For $Q_1 > Q_0$, evaluation at the positive minimizer r_1 for Q_1 gives $J_d(Q_1) - J_d(Q_0) \ge d(Q_1 - Q_0)r_1^2 > 0$. □

3 A positive energy gap uniform over hidden width

The next lemma addresses a subtlety of the online order: only the first new label of a block has been revealed when its error is counted. That intermediate history need not have the covariance certificate.

Lemma 3 (Uniform gap to a different label)**.** *Suppose a history satisfies Theorem 2, its affine minimizer a has nonzero scores on the entire cube, and its scalar output predicts the wrong label at a query x_0. If some globally margin-realizable function fits the history and the correct query label, there exists $e_1 > J_d(Q)$ such that every network fitting those constraints and global margins has energy at least e_1, at every width.*

Proof. We use a closed relaxation that includes all finite-width networks and all limits of their neuron distributions. A relaxed neuron is (u, b, H), where $H \in [-1, 1]^X$ and

$$H(x) = \operatorname{sign}(u \cdot \bar{x}) \quad \text{whenever } u \cdot \bar{x} \ne 0.$$

This is a closed subset of a finite-dimensional Euclidean space. Its output is the continuous function of the relaxed coordinates

$$b\big[\tanh(u\cdot\bar{x})+\epsilon H(x)\big].$$

At zero preactivation the relaxation permits any jump value in $[-\epsilon,\epsilon]$. The proof of Theorem 2 still applies: its only use of the jump term in (9) was the bound on its absolute average by ϵ.

Consider all probability measures on relaxed neurons satisfying the closed training margins and the correct signed query margin at least $1/2$. We may omit global constraints at other inputs, which only enlarges this class. It is nonempty by hypothesis. The infimum of mean energy $\mathbb{E}[d\|u\|^2+b^2]$ is attained. To see this, take a minimizing sequence with a common finite energy bound. The second moments make it tight, while the H coordinates lie in a compact set. A weakly convergent subsequence is supported on the same closed relaxed-neuron space. The energy is lower semicontinuous. Each output constraint passes to the limit: its integrand has magnitude at most $M|b|$, and the second-moment bound makes these integrands uniformly integrable. Truncating b gives bounded continuous test functions, which proves the required convergence of means.

Theorem 2 puts the attained energy at least $J_d(Q)$. If equality held, its atomwise equality proof would force almost every neuron to have $u=\pm ra$ and the matching readout. These rows have no zero preactivation anywhere on the cube. All their jump values are therefore forced, and every neuron has the same scalar output, which is wrong at x_0. This contradicts the correct query margin. The attained energy is consequently some $e_1>J_d(Q)$. Every finite-width network is a uniform empirical measure in this relaxation, proving the width-uniform claim. □

4 A selector family with short covariance certificates

Fix $k\ge 8$, put $M=2^k$ and $d=2k+2$, and use Boolean inputs $\xi=(x,y,z,s)\in\{-1,1\}^{2k+2}$. The two k-bit blocks encode

$$X=\sum_{i=0}^{k-1}2^i\frac{x_i+1}{2},\qquad Y=\sum_{i=0}^{k-1}2^i\frac{y_i+1}{2}.$$

For $0\le t\le M-1$, define the homogeneous affine scores

$$A_t(\xi)=-ts-(t+1)z+\sum_{i=0}^{k-1}2^i(x_i-y_i),\qquad Q_t=\|A_t\|_2^2=\frac{2(M^2-1)}{3}+2t^2+2t+1. \tag{12}$$

Here $\|A_t\|_2$ denotes the Euclidean norm of its coefficient vector. Every score is an odd integer on the full cube. On $s=1,z=-1$ it is $1+2(X-Y)$, and on $s=z=1$ it is $2(X-Y)-2t-1$. The target is

$$f(\xi)=\operatorname{sign}A_{M-1}(\xi). \tag{13}$$

It is odd. On $s=1$, it is positive exactly when $z=-1$ and $X\ge Y$.

4.1 Small pairwise-independent free-bit sets

For $r\ge 1$, let $\ell=\lceil\log_2(r+1)\rceil$ and choose r distinct nonzero vectors $v_1,\dots,v_r\in\mathbb{F}_2^\ell$; the first r nonzero binary vectors in their usual integer order will do. Let

$$\mathcal{P}_r=\Big\{\big(-(-1)^{v_j\cdot\omega}\big)_{j=1}^r:\omega\in\mathbb{F}_2^\ell\Big\}.$$

Use the uniform seed law, and let $\mathcal{P}_0$ consist of the single empty assignment. The chosen vectors span $\mathbb{F}_2^\ell$, so the seed map is injective. Thus

$$n_0:=|\mathcal{P}_0|=1,\qquad n_r:=|\mathcal{P}_r|=2^{\lceil\log_2(r+1)\rceil}\le 2r\quad(r\ge 1). \tag{14}$$

Each free bit has mean zero, and distinct free bits have product mean zero, because v_i, v_j, and $v_i + v_j$ are nonzero. The all-minus assignment belongs to $\mathcal{P}_r$, from the zero seed. Consequently these sets reproduce every first and second moment of independent uniform free bits, while using only $O(r)$ points.

Define the following groups, all on $s = 1$.

1. The positive diagonal group $\mathcal{D}$ has $z = -1$, $x = y$, with the k equal bits ranging over $\mathcal{P}_k$.
2. For each $i = 0, \dots, k-1$, the negative carry group $\mathcal{C}_i$ has $z = -1$,
$$x_j = 1,\ y_j = -1\ (j < i), \qquad x_i = -1,\ y_i = 1,$$
and its $k - 1 - i$ upper equal pairs range over $\mathcal{P}_{k-1-i}$. Thus $X - Y = -1$.
3. For $k \le t \le M - 1$, let $b_i = \mathrm{bit}_i(t)$. The negative query group $\mathcal{G}_t$ has $z = 1$,
$$(x_i, y_i) = (1, -1) \quad \text{if } b_i = 1,$$
and its remaining equal pairs range over $\mathcal{P}_{k-\mathrm{popcount}(t)}$. Hence $X - Y = t$. The point q_t with $X = t, Y = 0, z = s = 1$ belongs to this group.

All stated labels agree with the target. With every labeled point (ξ, y) also include its antipode $(-\xi, -y)$. Let H_t be the history containing the paired groups $\mathcal{D}$, all $\mathcal{C}_i$, and $\mathcal{G}_k, \dots, \mathcal{G}_t$.

Lemma 4 (Affine minimum and covariance certificate)**.** *For every $k \ge 8$ and $k \le t \le M - 1$, the unique minimum-norm affine unit-margin vector fitting H_t is $(0, A_t)$, where the first coordinate is the coefficient of the appended constant. It has squared norm Q_t and absolute unit margin on the whole cube. It admits an active KKT probability distribution whose signed augmented inputs Z satisfy*
$$\mathbb{E}Z = (0, A_t)/Q_t, \qquad Z \cdot (0, A_t) = 1, \qquad \mathbb{E}ZZ^{\mathsf{T}} \preceq 2I.$$

Proof. Write
$$T = t + \tfrac{1}{2}, \qquad B = \frac{M^2 - 1}{3}, \qquad Q = Q_t = 2B + 2T^2 + \tfrac{1}{2},$$
and set
$$c_i = 2^i + Tb_i, \qquad S_i = \sum_{j=i}^{k-1} 2^{j-i} c_j, \qquad S_k = 0, \qquad \mu_i = c_i + S_{i+1}.$$
These positive numbers obey
$$S_i = \sum_{r=i}^{k-1} \mu_r, \qquad S_0 = B + Tt = \frac{Q - T - 1/2}{2}.$$
Give $\mathcal{D}$ total KKT weight $P = S_0 + 1/2$, each $\mathcal{C}_i$ total weight μ_i, and $\mathcal{G}_t$ total weight T. Within each group distribute the weight according to its uniform free-bit law. These weights sum to Q. Every point in their support has signed A_t-margin one.

For the moment calculations use orthonormal coordinates
$$d'_i = \frac{x_i - y_i}{\sqrt{2}}, \qquad h'_i = \frac{x_i + y_i}{\sqrt{2}}, \qquad u = \frac{s - z}{\sqrt{2}}, \qquad v = \frac{s + z}{\sqrt{2}}.$$
The coefficient vector of A_t has components
$$(\sqrt{2}\, 2^i)_i, \quad (0)_i, \quad 1/\sqrt{2}, \quad -\sqrt{2}T.$$

The signed KKT weighted mean of the points has these same components:

$$\sqrt{2}(\mu_i - S_{i+1} - Tb_i) = \sqrt{2}\,2^i, \qquad \sqrt{2}(P - S_0) = 1/\sqrt{2}, \qquad -\sqrt{2}T.$$

The free-bit means vanish. This proves stationarity for the homogeneous coefficient vector. Split every point's weight equally between it and its labeled antipode. In the augmented coordinates this cancels the constant-coordinate mean and leaves the other signed means unchanged. Hence $(0, A_t)$ has a positive active KKT certificate and is the unique minimum of the strictly convex squared norm. It fits all earlier $\mathcal{G}_j$ because its score there is $2j - 2t - 1 \le -1$.

It remains to bound the unsigned covariance $C = \mathbb{E}\xi\xi^{\mathsf{T}}$ of the normalized weights. Pairwise independence of the free equal bits makes the h' block diagonal with entries at most two; it also makes all its cross terms with the other coordinates zero. Put $\delta = T/Q$. The remaining block, in the order (d', u, v), is

$$C_0 = \begin{pmatrix} K & h_u & h_v \\ h_u^{\mathsf{T}} & 2(1-\delta) & 0 \\ h_v^{\mathsf{T}} & 0 & 2\delta \end{pmatrix}, \tag{15}$$

where

$$h_u = -2c/Q, \qquad h_v = 2\delta b,$$
$$K_{ii} = 2S_i/Q + 2\delta b_i, \qquad K_{ij} = -\frac{2}{Q}\left[2^j + Tb_j(1-b_i)\right] \quad (i < j).$$

Only first and second free-bit moments enter these formulas.

Since $T < M$, $Q \ge M^2/2$, and $S_i \le 2^{-i}S_0 < Q/2^{i+1}$,

$$\delta \le 2/M, \qquad \|K\|_{\mathrm{op}} \le 1 + 6k/M.$$

For the latter estimate, each diagonal entry is at most $1 + 4/M$ and each off-diagonal absolute value is at most $6/M$; apply Gershgorin. Furthermore, $\sum_i c_i^2 \le 2B + 2T^2k$, $B/Q \le 1/2$, and $T \ge k$ give

$$\frac{\|h_u\|^2}{2\delta} \le \frac{2}{k} + \frac{8k}{M}, \qquad \frac{\|h_v\|^2}{2(1-\delta)} \le \frac{16k}{M^2}.$$

The Schur complement of the last two coordinates in $2I - C_0$ is

$$2I - K - \frac{h_u h_u^{\mathsf{T}}}{2\delta} - \frac{h_v h_v^{\mathsf{T}}}{2(1-\delta)}.$$

It is positive definite because

$$1 + \frac{2}{k} + \frac{14k}{2^k} + \frac{16k}{4^k} < 2 \qquad (k \ge 8). \tag{16}$$

At $k = 8$ the three positive terms after one sum to less than 0.69, and each decreases thereafter. Thus $C \preceq 2I$. Pairing the augmented inputs makes their covariance $\mathrm{diag}(1, C)$, completing the certificate. □

4.2 The fixed order and the bounded teacher

Start the input permutation with H_k, in any fixed order. At stage $t = k+1, \dots, M-1$, present q_t first, then its antipode, then the remaining points of $\mathcal{G}_t$ and their antipodes in a fixed order. At the end of the stage the history is H_t. Different groups are distinguished by selector face, value of $X - Y$, or highest differing bit, and the antipodes have $s = -1$, so there are no repeated inputs. Append all remaining cube points in any fixed order.

From (14), the initial calibration has at most

$$2\left(n_k + \sum_{r=0}^{k-1} n_r + n_{k-\mathrm{popcount}(k)}\right) \le 2k^2 + 6k + 2 \tag{17}$$

points. Each subsequent stage contains at most $4k$ inputs, including its designated first query.

Before q_t, Lemma 4 gives the affine minimizer A_{t-1}. Its score at q_t is $+1$, whereas the target label is negative. The exact profile and unique-function conclusion of Theorem 2, together with the posterior comparison below, therefore give a cold-posterior mistake on every designated first query, provided

$$d = 2k + 2 \ge 4\max\{1, \epsilon^{-6}\}.$$

There are

$$M - k - 1 = 2^k - k - 1 \ge 2^{k-1} \tag{18}$$

such queries. The unique minimizing function at the preceding history handles all competitors, including functions that would assign the same label to q_t and its as-yet-unrevealed antipode. The posterior argument thus does not assume that the learner's function is odd.

The target has Boolean circuit size $\Theta(k)$. A comparison circuit computes $X \ge Y$ with $O(k)$ bounded-fan-in gates, followed by the selector conjunction. Its odd extension on $s = -1$ is computed by negating the original inputs conditionally on s and multiplying the output sign by s, again using $O(k)$ gates. Conversely, all $2k$ main bits are essential, which gives the matching bounded-fan-in lower bound.

Finally set $C_\epsilon = \lceil 1/\epsilon \rceil$. The vector A_{M-1}/M has every coefficient of magnitude at most one, and has nonzero score at every input. Use C_ϵ identical hidden neurons with this incoming vector, zero appended-input bias, and readout weights one. Their signed output on the target is strictly greater than $C_\epsilon \epsilon \ge 1$. This is a teacher of the same depth, width C_ϵ independent of k, and individual weights at most one. In particular the online result applies at every learner width above this bounded teacher complexity, as well as at every other positive width covered by the exact-profile theorem.

5 From the energy gap to the actual posterior

We record the Gaussian estimate used for both online and IID learning. It also handles the activation's discontinuity.

Lemma 5 (A product-box comparison)**.** *Suppose a fixed history has a cloned single-neuron witness with energy e_0, global weak output margins, and nonzero preactivation at every cube input. Suppose every network in a competing event has energy at least $e_1 > e_0$, at every width. Then there exist $\Delta > 0$ and $C \ge 0$, independent of N, σ, such that the competing posterior mass is at most*

$$\exp[-\Delta/(2\sigma^2) + CN] \tag{19}$$

whenever $(d+2)N\sigma^2 < e_1$. The posterior conditions on the history and the global output margin.

Proof. Put $b_j = Nv_j$ and use row coordinates $\theta_j = (\sqrt{d}\,u_j, b_j) \in \mathbb{R}^p$, where $p = d + 2$. These are independent centered Gaussians with covariance $s^2 I_p$, where $s^2 = N\sigma^2$, and $E_N = N^{-1}\sum_j \|\theta_j\|^2$. Slightly amplify the witness readout, keeping its energy below e_1 and making every output margin strict. Its preactivations stay away from zero on the finite cube. A sufficiently small fixed open coordinate box B about this row therefore consists entirely of neurons with the same strict full-cube margin function, and satisfies

$$V := \mathrm{Vol}(B) > 0, \qquad R_B := \sup_{\theta \in B} \|\theta\|^2 < e_1.$$

If every row lies in B, their average has those same margins. The posterior denominator is consequently at least

$$\left[V(2\pi s^2)^{-p/2} e^{-R_B/(2s^2)}\right]^N.$$

For the numerator, NE_N/s^2 is chi-square with Np degrees of freedom. The exponential-moment bound, valid when $e_1 > ps^2$, gives

$$\log \mathbb{P}(E_N \geq e_1) \leq -\frac{Ne_1}{2s^2} + \frac{Np}{2}\left[1 + \log \frac{e_1}{ps^2}\right].$$

The logarithmic s^2 terms cancel upon division by the denominator. The log ratio is at most

$$-\frac{e_1 - R_B}{2\sigma^2} + N\left\{\frac{p}{2}\left[1 + \log \frac{2\pi e_1}{p}\right] - \log V\right\}.$$

Take $\Delta = e_1 - R_B > 0$ and replace the expression in braces by its maximum with zero. □

At every completed selector stage, the affine certificate in Lemma 4 satisfies Theorem 2, and its affine scores are nonzero on the full cube. Its scalar output has the wrong sign at the next designated query. The target is a globally realizable correct competitor, so Lemma 3 supplies an energy gap uniform in N. Lemma 5 applies to the event that the next label is predicted correctly. This step does not require the incomplete next history to have a covariance certificate.

For each fixed N, (19) tends to zero as $\sigma \downarrow 0$. Under (2) it becomes

$$\exp[-\Delta N^\gamma/(2\tau^2) + CN],$$

which tends to zero for every fixed $\gamma > 1$, $\tau > 0$. The tail condition also holds eventually. For $\gamma \geq 1$, every $N \geq 1$ satisfies both the tail condition and a desired competing-mass bound δ/T_k if

$$\tau^2 < \frac{e_1}{d+2}, \qquad \frac{\Delta}{2\tau^2} - C \geq \log(T_k/\delta).$$

Take common constants over the finite list of designated histories. A union bound proves the simultaneous sampling statement in Theorem 1. The same probability estimates imply its posterior-majority and confidence statements. No sampling independence is needed for this union bound.

6 Energy compression and polynomial IID learning

The IID argument uses only boundedness and monotonicity of the activation. It does not require that all low-energy output functions be affine thresholds. A bounded monotone neuron is an average of threshold neurons; sampling from that average gives a finite class containing every low-energy full-margin sign function.

Lemma 6 (Compression into a majority of thresholds)**.** *Let $\phi : \mathbb{R} \to [-M, M]$ be nondecreasing, where $M > 0$, and let $E \geq 0$. Consider all shallow networks on $\{-1, 1\}^d$, over all finite widths, whose energy (5) is at most E and whose outputs have absolute margin at least $1/2$ everywhere. Every resulting Boolean function is an unweighted majority of*

$$m_* = \left\lceil 256(E+1)M^2(d+1) \right\rceil$$

affine threshold functions. Consequently the class $\mathcal{G}_E$ of these Boolean functions satisfies

$$\log |\mathcal{G}_E| \leq m_*(d+1)^2 \log 2. \tag{20}$$

Ties in the majority may be resolved by any fixed convention.

Proof. Write $F(x) = \sum_j v_j \phi(u_j \cdot \bar{x})$ and $B = \sum_j |v_j|$. Cauchy–Schwarz gives

$$B^2 \leq N \sum_j v_j^2 \leq E.$$

The full margin implies $B > 0$. For a fixed neuron, choose U uniformly from $[0, 1]$ and define

$$h_U(x) = \begin{cases} +1, & U \leq [\phi(u_j \cdot \bar{x}) + M]/(2M), \\ -1, & U > [\phi(u_j \cdot \bar{x}) + M]/(2M). \end{cases}$$

Then $\mathbb{E}_U h_U(x) = \phi(u_j \cdot \bar{x})/M$. Monotonicity means that every such upper level set is a threshold in $u_j \cdot \bar{x}$. On the finite cube, almost every U avoids all the finitely many activation values used here. Its truth table is therefore a strict affine threshold function, including a possible constant table. Indeed one can place an affine threshold between the largest negative score and the smallest positive score. This argument allows jumps in ϕ and requires no continuity or parameter rounding.

Now select neuron j with probability $|v_j|/B$, sample its level U, and multiply its threshold table by $\operatorname{sign}(v_j)$. Negating a strict threshold table gives another strict threshold table. Thus there is a probability distribution over affine threshold functions h such that

$$F(x) = MB\, \mathbb{E} h(x).$$

Take m_* independent draws $h_1, \ldots, h_{m_*}$. Hoeffding's inequality and a union bound over the cube give

$$\mathbb{P}\left(\max_x \left| \frac{MB}{m_*} \sum_{j=1}^{m_*} h_j(x) - F(x) \right| \geq \frac{1}{8} \right) \leq 2^{d+1} \exp\left(-\frac{m_*}{128 M^2 B^2} \right) < 1.$$

For the last inequality, use $B^2 \leq E$ and the definition of m_*; when $E = 0$ the full-margin class is empty. There is therefore a realization with uniform error less than $1/8$. Its sum has the same sign as F at every input and has no zero value. Multiplication by the positive scalar MB/m_* does not affect that sign, proving the majority representation.

Let $\mathcal{H}_d$ denote the affine threshold truth tables on the cube. The 2^d input points determine hyperplanes in a $(d+1)$-dimensional coefficient space. Adding one hyperplane at a time gives the usual region bound

$$|\mathcal{H}_d| \leq \sum_{j=0}^{d+1} \binom{2^d}{j} \leq (2^d + 1)^{d+1} \leq 2^{(d+1)^2}.$$

There are at most $|\mathcal{H}_d|^{m_*}$ ordered threshold lists of length m_*. This proves (20). □

Theorem 7 (IID guarantee)**.** *Fix $\epsilon > 0$, an affine threshold target f on the cube, an input distribution $\mathcal{D}$, and $\eta, \beta \in (0,1)$. Put*

$$E_\epsilon = \epsilon^{-2}, \qquad M_\epsilon = 1 + \epsilon, \qquad m_\epsilon = \left\lceil 256(E_\epsilon + 1)M_\epsilon^2(d+1) \right\rceil, \qquad \Lambda_\epsilon = m_\epsilon(d+1)^2 \log 2.$$

If the number of IID training examples satisfies

$$n \geq \frac{4}{\eta}\left[\Lambda_\epsilon + \log(1/\beta)\right], \tag{21}$$

then, with sample probability at least $1 - \beta$, the posterior correct-label probability is greater than $2/3$ outside a set of $\mathcal{D}$-mass at most η in each of these regimes:

1. *at every fixed width, for all sufficiently small $\sigma > 0$;*
2. *under (2), at every sufficiently large width for any fixed $\gamma > 1$ and $\tau > 0$;*
3. *at every width and every $\gamma \geq 1$ under (2), for every sufficiently small fixed $\tau > 0$.*

The thresholds can be chosen independently of the realized sample and $\mathcal{D}$. At fixed ϵ, the sample bound is $O_\epsilon((d^3 + \log(1/\beta))/\eta)$.

Proof. Choose any affine separator a with $f(x)a \cdot \bar{x} \geq 1$ on the whole cube, and let $Q = \|a\|^2$. Cloning $u_j = ra$ and $b_j = 1/[2(\epsilon + \tanh r)]$ gives a full-target weak-margin witness of energy

$$dQr^2 + \frac{1}{4(\epsilon + \tanh r)^2}.$$

Its derivative at $r = 0$ is negative, so a sufficiently small positive r gives energy $e_f < 1/(4\epsilon^2) < E_\epsilon$. Every preactivation is nonzero. This upper bound needs no covariance certificate for the target. Apply Lemma 5 to this witness and the event $E_N \geq E_\epsilon$. The witness fits every target-consistent history, so the resulting posterior bound is uniform over all such histories. It tends to zero in the first two regimes and can be made at most $\eta/12$ at every width in the third.

Lemma 6 gives a single class $\mathcal{G}_{E_\epsilon}$, independent of learner width, containing every full-margin function represented below that energy threshold. Its logarithmic size is at most Λ_ϵ. A consistency union bound and (21) imply, with probability at least $1 - \beta$, that every sample-consistent member of this class has population error at most $\eta/4$. On this event, the posterior mean population error is at most $\eta/4 + \eta/12 = \eta/3$. An input with correct-label posterior probability at most $2/3$ has wrong-label posterior probability at least $1/3$. Markov's inequality therefore bounds the mass of these inputs by η, proving the claim. □

7 What the small jump changes

For the histories in the construction, the jump supplies a signed activation of size ϵ even at arbitrarily small nonzero preactivation. The tanh component then makes a positive incoming scale strictly preferable to zero. The resulting energy is a strictly increasing function of the affine margin norm, despite remaining below $1/(4\epsilon^2)$. The covariance certificate prevents a nonlinear mixture of neurons from improving that energy.

The online error count does not come from computational limitations or a favorable selection of one tied predictor. A positive energy gap separates the actual wrong scalar output from every network that would give the correct next label, uniformly over hidden width. The same prior

nevertheless gives polynomial IID sample complexity. The fixed input order, together with exact global-margin conditioning, is what distinguishes the two learning guarantees.

This is a theorem about a discontinuous activation uniformly close to tanh. It is not a theorem about tanh itself, and it does not establish stability under replacing the jump by a continuous transition. The corresponding all-width question for ordinary tanh, ReLU, and hardtanh under the same prior remains separate.